

communication options at the places where something is not already present is lost if you need power. So that is where the low-power local area network (LAN) and low-power wide area network (WAN) are becoming a reality. These are all essentially the need of the hour.

This is why LoRa and NB-IoT are being picked up. Each of them has its own benefits. The big benefit of LoRa is that it is non-regulated, long ranged, and requires very minimal power. Anyone can buy an off-the-shelf LoRa gateway and put their own network server and then connect.

Another factor to consider is market reach. If you are playing in a specific segment, it is important to know how you are going to reach out to the market, what is the market demand, and how are you going to enable it. These are some of the key parameters.

There are many protocols that have a high bandwidth and there are many others that provide a speed of a few kbps only. Bluetooth Low Energy (BLE) and Bluetooth, which have a higher bandwidth, allow you to send video files and PowerPoint presentations, etc.

But as soon as you add LE, it will restrict the bandwidth while increasing battery life and range. If your device cannot communicate to the external world with a significant interoperability, then you are restricting yourself. Consumer IoT may not need to be interoperable since it can easily be adapted to any other technology.

Then comes the cost and affordability that define the market you are targeting and the business problems that are being solved. Because, at the end of the day, if any of the TSP, SI, CSP goes to a customer then the customer only seeks the return on investment (RoI) from them on the overall solution. Mostly the deals are funded by these people only.

The typical approach is to minimise the cost of a third party and improve your own margin. So, if a device costs too much, people may

start looking for alternate options. It is therefore important to align cost parameters to the use case.

The last but the most important part is modularity from a communication standpoint. While we have so many technologies available, nobody can guarantee if all these technologies will last for more than five years; with newer technologies coming out the old ones become obsolete.

Some important use cases

It is important to check if the communication that we are performing is adding to the radiation or not. We all know that there is a continuous signal available in the environment for different options, but the convergence happens on the receiving device. The convergence strength increases at the receiving end. This is why it is advised not to keep your phone in your shirt's pocket.

Let us consider some important use cases of 5G.

Healthcare and hospitals. Anything that deteriorates a patient's health is not desirable. The industry that you want to cater to should be considered in terms of wireless protocol you would be using. Most of the healthcare devices have to go through health and safety clearances and other similar certifications.

Workplaces. Most workplaces have Wi-Fi connectivity. But if you need an occupancy sensor on every desk then Wi-Fi may not be the best option. This is because those work desks are already configured for a particular number of ports on each desk with their utility assigned. There could be wires hanging around all over the place out of their intended sockets, so here LoRa may be a great option.

LoRa based products are gaining significant adoption. The Covid pandemic has proven that there is no need for the entire staff to be present in an office to work. You can optimise your work better in a hybrid work setting where some people work from home

or anywhere. This is where market assessment plays an important role.

Predictive maintenance. Predictive maintenance does not only happen on an application. An application is just one layer which runs the algorithm. It is the device's duty to decide whether the right data has been fed to that algorithm in a timely manner, as many people miss out on the time synchronicity of the data.

If this data is not on time sync then your predictions cannot be accurate. Depending upon how out of sync the data is, the accuracy level will go down as well. That is why we talk about predictive maintenance a lot. If you are building a sensor set for predictive maintenance use cases, then it is better that you use the same communication technology with similar time lag which is going to come from the system connection.

5G is going to combine different layers that are disruptive in nature. It can do broadband communication, high reliability communication, and much more. It also serves machines that are required in saving lives. Anything that requires a long range, lower bandwidth, and a high speed will be catered to by 5G. The reality today is that there are a number of devices and sensors which support 5G and people are mostly focusing only on broadband now.

Finally, this is the right time to invest in 5G as it is becoming a reality. India is adapting to this change very quickly. Private 5G is going to be faster than public 5G from an IoT use case perspective. It is going to ultimately address the basic concern of plants' operation on issues like security, connectivity, reliability, and multiple communication requirements through a single technology. **EFY**

This article is based on a tech talk given by Ashish Khare, General Manager, Global Head IoT and 5G, Wipro Limited at India Electronics Week 2022. It has been transcribed and curated by Laveesh Kocher, a tech enthusiast at EFY, who has a knack for open source exploration and research